Amos Kipchirchir

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Education

Kirinyaga University

Bachelor of Science in Software Engineering Graduation 2026

Relevant Coursework: Object-Oriented Analysis & Design, Database Management, Client-Server Architecture, HTTP & Web Systems

Experience

Machine Learning

Developed machine learning models using real-world datasets to generate predictive insights, applying data preprocessing, feature engineering, and model optimization techniques. Structured and cleaned multi-source data to improve quality and usability, enabling more reliable model performance. Deployed models through backend APIs to support integration with applications and real-time usage

Web Development

Developed responsive web applications using modern frameworks, focusing on clean design and user experience. Implemented backend functionality and RESTful APIs to handle data processing and client-server communication Integrated databases and external services to build scalable and maintainable web solutions

Agents & Integration

Developed and integrated AI-powered agents into web applications, enabling conversational interfaces and task automation. Implemented chatbot functionality using language model APIs to enhance user interaction and accessibility. Connected AI services with backend systems to support real-time responses and seamless user experiences

Leadership & Activities

- Applied object-oriented design principles in academic and personal projects
- Built backend systems using Python and PHP, implementing client-server architecture
- Utilized HTTP protocols and database systems in web-based applications
- Solved algorithmic problems using efficient data structures and problem decomposition

Skills & Interests

Technical: Python (Advanced), C++ (Intermediate), PHP (Intermediate), SQL, FastAPI, REST APIs, Scikit-learn, Pandas, NumPy

Machine Learning: Model Training & Evaluation, Feature Engineering, Data Preprocessing, PyTorch/TensorFlow (Foundational), LLM Integration

Tools & Platforms: Git, GitHub, Docker, AWS (EC2, S3), PostgreSQL, GitHub Actions (CI/CD)

Concepts: Data Structures & Algorithms, OOP, System Design Basics, HTTP Protocols

Interests: Machine Learning Applications, Game Development, Algorithmic Problem Solving